Final Pick: Procedure Drift Finder vs Teach-Back Companion

For the team, 10 Oct 2026. Scores are judgment calls based on public web research, not product trials. Please verify before using any claim in the pitch.

Recommendation

Go with Procedure Drift Finder (Intelligent Industry track).

Why:

- **Clearer market gap.** The big players in this space watch workers with cameras. Nobody I found compares what workers actually write and say against the written procedure, step by step.
- **Less direct overlap with a giant.** Teach-Back Companion sits right next to a well-funded leader (Hippocratic AI). Drift Finder does not.
- **Cleaner demo and proof.** We plant 25 known "drifts" in synthetic data and show how many the tool catches, with evidence for each. No human subjects needed.
- **Fewer compliance hurdles.** The Health track has strict safety rules and we would want real-people testing. Industry lets us use synthetic data safely.
- **Likely less crowded.** Past AMD hackathon top entries included several clinical AI projects, so I expect the Health track to be busy. I cannot see ACT III submission counts yet, so this is an expectation, not a fact.

Choose Teach-Back instead only if the team has a real language or dialect edge and a hospital or clinic contact, and is willing to run a real-people test.

Side by side

Criterion	Procedure Drift Finder	Teach-Back Companion
Market gap clarity	Clear (leaders are camera-based)	Moderate (leader already reviews instructions by phone)
Direct overlap with a leader	Low	High (Hippocratic AI, Memora Health)

Criterion	Procedure Drift Finder	Teach-Back Companion
Likely track crowding	Lower	Higher
Feasibility for beginners	~7/10	~8/10
Needs real people or real data	No, synthetic data is enough	Yes, for credibility
Safety and compliance burden	Low	High (medical safety rules)
Evidence the problem is real	Strong (accident inquiries, studies)	Very strong (clinical trials)
Demo strength	Strong (planted drifts, evidence quotes)	Strong (live conversation)
Originality (my estimate)	~7 to 7.5/10	~7/10

Neither is an 8+ on pure originality. Do not claim "first" for either.

Idea 1: Procedure Drift Finder

What it is: A tool that reads a plant's written procedures (SOPs) and the text it already produces (shift handover notes, near-miss reports, maintenance logs), then flags where real practice differs from the procedure. Each flag shows the SOP step, the exact quote as evidence, how often it happens and on which shift. A person approves or rejects each flag.

Why it matters: Safety science calls this the gap between work-as-imagined and work-as-done. One study of 524 procedural steps in a high-risk industry found 34% showed discrepancies. Major accident inquiries (Piper Alpha, an LPG explosion) point to procedures that were routinely bypassed, a pattern called normalization of deviation.

Market gaps vs the leaders

Leaders: Intenseye and Protex AI (camera-based safety AI), plus EHS suites such as VelocityEHS, Cority and Intelex, plus newer generative assistants.

1. **Cameras and surveillance.** The leaders detect unsafe acts through CCTV. Watching workers carries heavy obligations in Europe (EU AI Act, works councils). Our tool reads text the plant already produces, so no new monitoring.
2. **Steps and intent.** One third-party review says camera tools struggle with behavior that needs context or intent, such as a worker about to bypass a lockout. Our tool compares the procedure step by step in text.
3. **Three versions of the rules.** An independent review notes sites often find the written procedure, what supervisors enforce and what actually happens are three different things. We make that gap visible.
4. **Detection that never becomes action.** Reviewers flag the seam where a detection either becomes an assigned corrective action or a dashboard nobody opens. We output a ranked list with suggested SOP updates for human approval.
5. **Price and access.** The leaders are enterprise, quote-only, often priced per camera. Our version uses an open model on AMD hardware and can run on a plant's own machines.
6. **Generative EHS assistants** mostly help log incidents and draft documents. I did not find one that analyzes patterns against the SOP, but I did not test them all.

What we build in 7 days (MVP)

- 3 to 5 synthetic SOPs (for example lockout-tagout, forklift charging, confined-space entry).
- About 200 synthetic handover notes and near-miss reports, with **25 planted drifts**.
- Output: ranked drift list with SOP step, evidence quote, severity and trend by shift.
- A review screen where a person approves or rejects each flag.
- To make test notes more realistic, base some on public incident narratives (for example OSHA's published severe injury reports; verify access).

Measurable results for judges

- Of 25 planted drifts, how many are found (recall) and how many flags are wrong (precision).
- Throughput: hundreds of notes processed in X seconds on an AMD MI300X.
- A slide titled "Where our AI is wrong" with real false positives.

Risks

- Synthetic data may look too clean. Mitigation: public incident narratives, plus messy, multi-language notes.
- False alarms. Mitigation: human approval step and an honest precision number.
- Tracks overlap: confirm the project fits Intelligent Industry (it does, as a safety and operations product).

Idea 2: Teach-Back Companion

What it is: After discharge instructions are given, the patient explains them back in their own words, in their language. The AI scores each instruction (dose, timing, warning signs) as understood, partly understood or wrong, and flags gaps to a nurse. It never gives diagnosis or treatment advice.

Why it matters: In one emergency department study, patients leaving with a comprehension deficit fell from 49% to 11.9% when teach-back was used. A meta-analysis found teach-back reduced readmission in heart-failure patients. A 2026 pilot found that translating instructions alone did not improve comprehension in most areas.

Market gaps vs the leaders

Leaders: Hippocratic AI (deployed with a large US hospital system for post-discharge calls) and Memora Health (text-based follow-up), plus hospital portals.

1. **Understanding vs satisfaction.** Published descriptions of the leaders emphasize check-in calls, reviewing medication instructions, symptom checks and patient ratings. I did not find per-instruction comprehension scoring. This is our best gap, but I could not verify their internals.
2. **Nurse-facing misunderstanding map.** Teach-back works but nurses do it by hand under time pressure. A dashboard showing who misunderstood which instruction saves that time.
3. **Language and dialect coverage.** Only a gap if the team can build and test dialect cases the leaders do not cover. Unknown until we know the team's languages.
4. **Privacy.** An open model on AMD hardware can run inside a hospital. I did not verify how the leaders host their models.

Risks

- **Direct overlap:** a judge may say "Hippocratic AI already does this". We would need a sharp answer about per-instruction scoring.
- **Safety rules:** the track forbids presenting autonomous diagnosis or treatment as advice. We must show how we handle wrong answers, sensitive data, uncertainty and human review.
- **Validation:** simulated patients are unreliable (research shows LLMs overestimate comprehension), so we would need 15 to 20 real people. That takes time and ethics care.
- **Crowding:** health is a popular hackathon category.

Honest originality statement for the team

Both ideas are about 7 out of 10. The strength is the **combination plus proof**: a clear user, a real gap, measured results and honest failure analysis. Past winners won on technical depth, measurable numbers and working demos, not on a slogan.

Decisions for the team

1. **Vote on the idea by tomorrow night** (Oct 11), so we can start on Oct 12.
2. **Tell us your languages:** which does the team speak, and does anyone know a plant, clinic or hospital?
3. **Assign roles:** lead, two builders, front-end, data and research, pitch and demo.
4. **Everyone signs up** for the AMD AI Developer Program and joins the team on the event page.
5. **Check the rules:** the exact submission deadline, the AMD requirement, and whether the Intelligent Industry track fits our final scope.